Substring Match

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	512 megabytes

Given two strings s and t, s is a string containing only lowercase letters, and t is a pattern containing lowercase and uppercase letters.

Defines that a string can be matched by a pattern:

- A lowercase letter in the pattern matches a corresponding lowercase letter.
- An uppercase letter in the pattern matches any number (including zero) of the corresponding lowercase letters.

For example, abb can be matched by aB; aaa can be matched by A; ac can be matched by aBc; ABC, $aABCc$, aab can not be matched by aB; aa can not be matched a.

Now you need to answer the length of the longest substring of s that can be matched by t.

Input

This problem contains multiple test cases.

The first line contains an integer $T(1 \leq T \leq 100)$ indicating the number of test cases.

For each test case, the first line contains a string $s(1 \leq |s| \leq 10^5)$, the second line contains a string $t(1 \leq |t| \leq 10^5)$.

It's guaranteed that $\sum |s| \leq 10^5$, $\sum |t| \leq 10^5$, and **total of uppercase letters of t no more than** 200 **for all test cases**.

Output

For each test case, output the length of the longest substring of s that can be matched by t in a line.

Example

standard input	standard output
3 aaa ABa aaa aAaBa aaa aa	3 3 2